

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>

High-level reasoning (concise):

- At birth , the brain ’s weight and size are not yet adult-like , with rapid post natal growth , particularly from the third trimester onward .
- The brain stem is critical for basic life functions (heart and respiratory control , circadian regulation , thermoregulation) and is relatively mature before birth . Its necessary circuits are largely complete , especially the upper brain stem nuclei .
- The diencephalon (which includes thalamus and hypothalamus) is largely formed before birth and is functionally active .
- The cerebellum is mostly developed before birth , contributing to early autonomic and motor control .
- The limbic system ’s nuclei are developed but not fully integrated with neocortical systems at birth .
- The hippocampus and amygdala show significant early growth and maturation .
- The basal ganglia are relatively mature early in gestation but also undergo substantial postnatal refinement .

In contrast , the cerebral cortex , particularly the prefrontal and higher-order areas such as the frontal and parietal lobes , is largely neurogenesis-complete before birth but postnatal synaptic refinement , pruning , and myelination continue through infancy and into childhood . This is why the frontal lobe is described as the most developmentally delayed region at birth .

Correct answer : G . cortex

</think>

<answer>

G . cortex

</answer>